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A Hybrid Machine Learning Approach Utilizing PCA and ICA Features for Stress Classification

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***Abstract*—Electroencephalographic (EEG) signal-based personal identification systems have notable advantages and disadvantages. These systems heavily rely on the stability of EEG signals, which several factors. One of the primary factors affecting the stability of EEG signals is an individual's emotional state. Among emotional states, stress significantly impairs people's ability to perform daily tasks. This research aims to identify stress levels, classified as low (2) and high (1), using features from Independent Component Analysis (ICA) and Principal Component Analysis (PCA). Machine learning methods, including Decision Tree, k-Nearest Neighbors (k-NN), Naive Bayes, Support Vector Machine (SVM), and Ensemble techniques, are employed to classify the stress levels. The dataset comprises 40 EEG recordings from the Stroop color-word test, and the data is split using a random holdout function with a ratio of 80% for training and 20% for testing. This study examines the most effective features for identifying stress levels and compares the performance of various machine learning models. The experimental results demonstrate that PCA is the most effective feature extraction method, achieving an average accuracy of 0.718 in stress level classification. Among the machine learning models tested, the Ensemble method performs the best, achieving an accuracy of 0.770 when using PCA features and 0.745 with ICA features. This study highlights the importance of selecting optimal features and machine learning techniques for improving stress detection in EEG-based systems. Further improvements in classification accuracy may be achieved by incorporating additional physiological signals or refining feature extraction techniques.**

Keywords—Stress classification; EEG; hybrid machine learning.

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I. INTRODUCTION

Since stress has both positive and negative impacts when managed effectively, and its effects can spiral out of control, research on stress has been conducted for a considerable amount of time. This is demonstrated by the fact that positive stress, or eustress, is known to provide goals and incentives to overcome obstacles and make individuals more resilient to similar problems in the future. However, the issue arises when concern and negative stress take hold, blocking the will to improve or overcome the obstacles in the way, thereby reducing the possibilities of what could have been accomplished. Electroencephalography (EEG) has been utilized in several studies to detect mental stress, particularly in conjunction with computer algorithms such as artificial intelligence and machine learning [1].

The action potential produced by neurons firing as a result of a chemical reaction in the brain is measured scientifically by an electroencephalogram. These action potentials will result in a potential difference, which is then calculated on a human scalp using electrodes. The use of brain waves as a biometric modality has garnered increasing attention in recent years. Brain waves are tough to fake and offer a great degree of uniqueness, permanence, and universality.

Current EEG-based personal identification systems either feed raw EEG data into deep learning models or employ features collected [2] from EEG signals, followed by a machine learning classifier [3]. With applications in neural engineering (such as brain-computer interfaces, or BCIs), neuroscience, and even business, electroencephalography (EEG) analysis has been a crucial tool in medicine. Machine learning has been employed by many of the analytical tools in EEG investigations to find pertinent data for neuroimaging and brain categorization [4]. Some machine learning methods that provide good results in classification or identification are SVM, K-NN [5], Naive Bayes [6], [7], [8], and Neural Network. It is crucial to develop a computer-aided diagnosis

(CAD) system that utilizes machine learning techniques to automatically distinguish between various types of EEG signals [9]. It is a continuous struggle to develop efficient methods for detecting stress in electroencephalogram (EEG) recordings.

Numerous research works have exhibited the relationships between EEG patterns and emotional states, such as anxiety, depression, and excitement. A stress analysis method based on EEG and the frontal alpha asymmetry index using the support vector machine (SVM) algorithm was presented in [10], [11]. To characterize the stress state using the EEG signal, a frequency domain-based analysis with a k-nearest neighbor (K-NN) algorithm was developed in [11], [12]. Deep learning techniques have also been used recently to categorize various mental states through the analysis of EEG signals. This research proposes identifying stress levels (low, high) based on ICA and PCA features using machine learning methods. The aim is to analyze the best features and machine learning methods in identification. Furthermore, this research chose to employ independent component analysis (ICA) and principal component analysis (PCA) because several studies have utilized them for feature extraction, which also facilitates information compression for faster processing.

In this research, Independent Component Analysis (ICA) and Principal Component Analysis (PCA) are applied as feature extraction techniques to improve the classification of stress levels from EEG data. ICA is used to decompose the EEG signals into statistically independent components, which helps isolate the underlying brain activity signals associated with stress. By separating these independent sources, ICA provides features that may capture more specific and relevant neural patterns. PCA, on the other hand, reduces the dimensionality of the EEG data by identifying the principal components that explain the most variance in the dataset. This method eliminates redundant information while retaining the most informative features. Both ICA and PCA are applied before classification to simplify the feature space and enhance the model's ability to distinguish between low and high stress levels. These techniques play a critical role in optimizing the data for machine learning models and improving classification accuracy.

II. Materials and Method

The Proposed Method section in the research describes the specific approach or techniques introduced by the authors to solve the problem at hand. This section typically explains the key components of the method, its structure, and how it differs from or improves upon existing methods. The explanation includes details on algorithms, models, data processing, and how the method is implemented in practice.

For example, if the research involves classifying stress levels from EEG signals using machine learning, the proposed method would describe how features are extracted from the EEG data (e.g., using PCA or ICA), the machine learning models used (e.g., Decision Tree, SVM, Ensemble), and the steps taken to optimize model performance. The method would also explain why these specific techniques were chosen, how the data was split for training and testing, and any unique aspects such as preprocessing techniques, parameter settings, or evaluation metrics. The proposed method aims to contribute something novel or improved to the field, whether through enhanced accuracy, increased speed, improved interpretability, or increased applicability.

The dataset used for stress classification consists of EEG recordings collected from participants during cognitive tasks such as the Stroop color-word test, which is known to induce stress by creating conflicts between word meaning and color perception. The data is labeled into two categories: low stress (2) and high stress (1). EEG signals were recorded using multiple electrodes, capturing neural activity associated with emotional and cognitive responses to stress. Feature extraction was performed using Independent Component Analysis (ICA) and Principal Component Analysis (PCA), followed by the application of machine learning models for classification. The dataset was split into training and testing sets using an 80/20 ratio for evaluation. For more information on similar datasets and methodologies, refer to studies in Frontiers in Neuroscience.

This research uses datasets from the study of cognitive processes under stress [13]. This research used EEG signal data related to the Stroop color-word test from trials 1, 2, and 3. The Stroop color-word test data identified low stress levels (scale value ≤ 5) and high stress levels (scale value >5). EEG data from 40 subjects identified stress levels using independent component analysis feature extraction (ICA) [34], [35], [36], and the principal component analysis (PCA) [37]. This research identifies stress levels using machine learning methods (Decision Tree (DT), K-NN (Nearest Neighbor), Naive Bayes, Support Vector Machine (SVM), and Ensemble). EEG data from 40 subjects were divided into training data and testing data randomly using the holdout function. The process of taking EEG data features with ICA involves taking 100 rows of data objects for each subject. The process of taking features with PCA is taking the principal component coefficients from each subject, resulting in 32 rows according to the 32-channel EEG column.

The dataset used in this research consists of EEG recordings collected from participants during cognitive tasks, specifically the Stroop color-word test, which aims to induce stress. To gather this data, participants are recruited and fitted with EEG electrodes placed according to the international 10-20 system, which allows for comprehensive monitoring of brain activity. While engaged in the task, the participants' EEG signals are recorded, capturing electrical activity from multiple brain regions. The dataset comprises raw EEG signals, demographic information such as age and gender, and stress level labels that categorize responses as either low or high, based on task performance and subjective stress assessments. Additionally, preprocessing steps are applied to clean the data by removing artifacts, followed by feature extraction techniques like Independent Component Analysis (ICA) and Principal Component Analysis (PCA) to enhance analysis. This structured dataset provides a foundation for applying machine learning algorithms to classify stress levels and investigate the relationship between cognitive demands and neural responses.

Independent Component Analysis (ICA) is a computational technique used to separate a multivariate signal into additive, independent components, making it particularly useful in the analysis of EEG data, where it helps isolate brain signals from noise and artifacts. The core principle of ICA is based on the assumption that the observed signals are linear

mixtures of non-Gaussian, statistically independent sources. Mathematically, if X represents the observed signals, and S is the independent source signals, then ICA aims to find a separating matrix W such that $Y = W \cdot X$, where Y contains the estimated independent components. The optimization problem involves maximizing the statistical independence of the components, often using measures like kurtosis or mutual information. ICA is widely applied in neuroscience for artifact removal, such as eye blinks and muscle activity, enhancing the quality of EEG analysis [38].

Principal Component Analysis (PCA) is a statistical technique used to reduce the dimensionality of data by transforming it into a new coordinate system where the most significant variance is captured in the first few principal components. In EEG data analysis, PCA helps in identifying the most significant features while reducing noise and redundancy. Mathematically, if X represents the data matrix, PCA finds a set of orthogonal vectors (principal components) by solving the eigenvalue decomposition problem of the covariance matrix $C = X^T X$, where the eigenvectors represent the principal directions and the corresponding eigenvalues indicate the amount of variance captured. The principal components are then ordered by their eigenvalues, and the first few components are retained for analysis, reducing the dataset's complexity while preserving its most important features. PCA is often used before applying machine learning models to improve classification performance by focusing on the most informative features [39].

The k-Nearest Neighbors (K-NN) algorithm is a simple, non-parametric machine learning technique used for classification and regression. It classifies a data point based on the majority class among its k-nearest neighbors in the feature space. Given a dataset with labeled points, when a new data point needs to be classified, the algorithm calculates the distance between this point and all other points in the dataset, typically using Euclidean distance

$$d(p,q) = \sqrt{\Sigma_{i=1}^{n}}(p_i q_i)^2 \quad (1)$$

where p and q are two points with n dimensions. The algorithm then selects the k closest points, and the most frequent class among them is assigned to the new point. K-NN is advantageous for its simplicity and effectiveness in well-behaved data but may struggle with high-dimensional spaces or imbalanced datasets. It is widely applied in pattern recognition, image processing, and EEG data classification [40].

A Decision Tree is a machine learning algorithm used for both classification and regression tasks. It works by splitting the dataset into subsets based on the value of input features to increase the purity of the resulting subsets. The structure of a decision tree consists of nodes that represent features, branches that represent decision rules, and leaves that represent the outcomes or class labels. The algorithm selects the best feature to split the data at each node by using a criterion such as Gini Impurity or Information Gain. For Gini Impurity, the formula

$$G = 1 - \sum_i^c p_i^2 \quad (2)$$

where p_i is the proportion of samples belonging to class i, and C is the number of classes. The tree grows until all the data is classified or a stopping condition is met. Decision Trees are easy to interpret and visualize, making them popular for understanding complex data structures, though they are prone to overfitting in noisy data [41].

Support Vector Machine (SVM) is a supervised machine learning algorithm primarily used for classification tasks. It works by finding the hyperplane that best separates data points of different classes in a high-dimensional space. The goal of SVM is to maximize the margin between the nearest data points from each class, called support vectors. Mathematically, given a dataset $\{(x_i,y_i)\}$ where x_i feature vectors and $y_i \in \{-1,1\}$ are class labels, the SVM finds a hyperplane defined as $w \cdot x+b=0$, where www is the normal vector to the hyperplane, and b is the bias. The optimization problem is to minimize $\frac{1}{2} ||w||^2$, subject to $y_i(w \cdot x_i+b) \geq 1$ for all $_i$, ensuring correct classification of the training data with maximum margin. For non-linearly separable data, SVM uses kernel functions (e.g., radial basis function) to map data into higher dimensions where it can be linearly separated. SVM is effective in high-dimensional spaces and widely used in fields like bioinformatics and EEG signal classification [42].

Naive Bayes is a probabilistic machine learning algorithm used for classification tasks, based on applying Bayes' Theorem with the assumption that all features are independent, hence "naive." Despite this simplifying assumption, it performs well in many applications, especially with large datasets. The formula for Bayes' Theorem is

$$PCX = \frac{P(X|C).P(C)}{P(X)} \quad (3)$$

where $P(X|C)$ is the posterior probability of class C given features X, $P(X|C)$ is the likelihood of the features given the class, $P(C)$ is the prior probability of the class, and $P(X)$ is the probability of the features. Naive Bayes works by calculating these probabilities for each class and selecting the class with the highest posterior probability as the prediction. It is computationally efficient and commonly used in text classification, spam filtering, and medical diagnosis due to its simplicity and effectiveness even when the independence assumption is not fully met.

This research identifies stress levels with classes low=2 (scale value ≤5), and high=1 (scale value >5) based on ICA and PCA features. The method of identifying stress levels uses machine learning (Decision Tree, K-NN, Naive Bayes, SVM, and Ensemble). Each trial of the EEG Stroop color-word test data is made into a machine learning and testing model by dividing using a random holdout function with a divisor value of 0.2. This research analyzes the best features in determining stress levels, and the best machine learning model. The proposal for this research is in Figure 1.

The stages of the entire research is shown in Figure 2. The initial EEG data for each subject was extracted with ICA and PCA features. Each subject was extracted with ICA, resulting in 100 rows, with a total of 4000 rows. Each subject was extracted with PCA resulting in 32 rows, with a total data of 1280 rows. Each trial of the EEG Stroop color-word test data has 4000 rows of data from the ICA feature and 1280 rows from the PCA feature. The data of each feature is divided into training and testing data using the holdout function with a divisor value of 0.2. The training data is used to create a machine learning model, and the model results are used to predict the test data. The prediction results are evaluated by

calculating the accuracy (Equation 1). TP (True Positive), TN (True Negative), FP (False Positive), FN (False Negative).

This research conducted several scenarios, to find the best machine learning features and models in identifying stress levels (low, high). The research scenarios are:

- EEG data is extracted using ICA and PCA features, and then a model is created tested, and evaluated.
- EEG data is extracted with ICA and PCA features, then the target class is balanced using the SMOTE method, and then a model is created tested, and evaluated.
- The EEG data was extracted using ICA and PCA features, then correlation calculations were performed to reduce the number of 32-channel columns to 16 columns, and then a model was created, tested, and evaluated.

The first scenario involved extracting EEG data using Independent Component Analysis (ICA) and Principal Component Analysis (PCA) features, followed by model creation, testing, and evaluation. In the second scenario, after feature extraction with ICA and PCA, the target class imbalance was addressed using the Synthetic Minority Over-sampling Technique (SMOTE) before creating and testing the model. The third scenario involved extracting EEG data with ICA and PCA and then performing correlation calculations to reduce the number of EEG channels from 32 to 16, after which the model was developed, tested, and evaluated. These scenarios aim to improve stress classification accuracy by exploring different preprocessing and data balancing techniques.

In the second scenario, EEG data is extracted using ICA and PCA features to identify key components relevant to stress levels. However, since the data may have an imbalanced target class (e.g., more high-stress cases than low-stress), the SMOTE (Synthetic Minority Over-sampling Technique) method is applied to balance the class distribution. SMOTE generates synthetic data for the minority class to prevent the model from being biased toward the majority class during training. After balancing the data, a machine-learning model is created, tested, and evaluated. This approach aims to improve the model's accuracy and generalization by ensuring it properly learns patterns from both classes, leading to more reliable classification results.

In the third scenario, the EEG data is extracted using ICA and PCA features to capture significant patterns related to stress levels. Correlation calculations are then applied to reduce the number of 32-channel columns to 16 columns, aiming to remove redundant or less informative channels. This step minimizes the dimensionality of the data, which can help improve computational efficiency and reduce the risk of overfitting in the machine learning model. By focusing on the most relevant features, this approach enhances the model's ability to generalize, ensuring that it performs well during the testing and evaluation phases. This method balances the trade-off between data complexity and classification performance.

The three scenarios outlined explore different techniques for improving the classification accuracy of stress levels using EEG data. In the first scenario, feature extraction through Independent Component Analysis (ICA) and Principal Component Analysis (PCA) was applied, followed by the creation, testing, and evaluation of the model. The second scenario expanded on this by addressing the class imbalance in the dataset using the Synthetic Minority Over-sampling Technique (SMOTE), which helped to balance the target classes and ensure more reliable classification results. This method prevented the model from favoring the majority class and enhanced overall accuracy. In the third scenario, after feature extraction with ICA and PCA, correlation calculations were performed to reduce the number of EEG channels from 32 to 16, optimizing data dimensionality. This reduction aimed to improve computational efficiency and minimize the risk of overfitting, while still focusing on the most relevant features. Each scenario demonstrated different preprocessing and balancing techniques, ultimately contributing to enhanced model performance and stress classification accuracy.

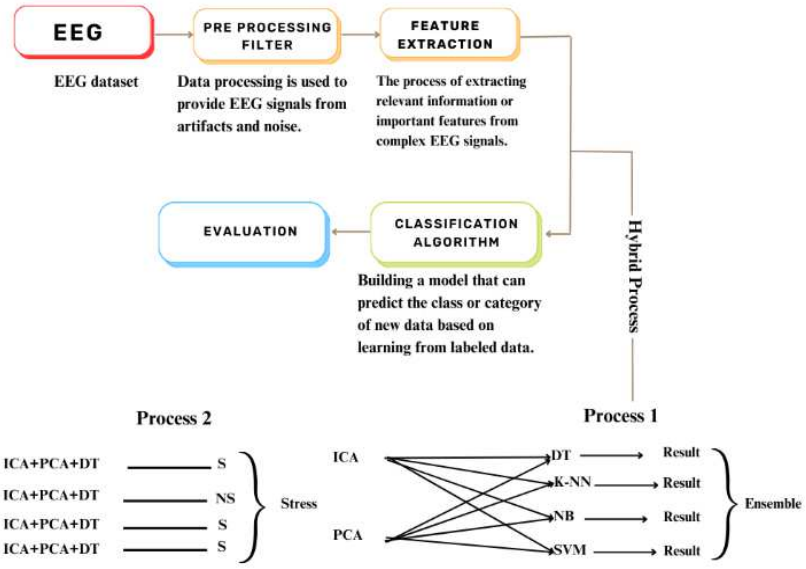

Fig. 1 Research proposal

The image illustrates the methodology for classifying stress levels using EEG (Electroencephalographic) data through machine learning techniques. It begins with the EEG dataset, which comprises raw signals collected from subjects. Before feature extraction, the data undergoes a pre-processing filter to eliminate noise and artifacts, ensuring high-quality signals

for subsequent analysis. For feature extraction, two methods, Independent Component Analysis (ICA) and Principal Component Analysis (PCA), are employed to distill relevant information from the complex EEG signals. Following this, various classification algorithms, including Decision Tree (DT), K-Nearest Neighbors (K-NN), Naive Bayes (NB), Support Vector Machine (SVM), and Ensemble methods, are applied to classify the extracted features. The image highlights a hybrid process where ICA and PCA features are processed independently through multiple classifiers, with results from each being compared. Additionally, it includes a specific scenario (Process 2) that combines CA + PCA + DT, differentiating between stress (S) and no stress (NS) levels based on the features extracted and classifier outputs. The final evaluation phase analyzes the classifiers' performance, assessing which combination of feature extraction techniques and classifiers yields the most effective stress classification results. Overall, this methodology emphasizes the systematic approach to processing EEG data and applying machine learning for accurate stress level classification.

The image emphasizes the significance of employing a hybrid approach to enhance classification accuracy by leveraging the strengths of multiple feature extraction techniques and classification algorithms. By utilizing both ICA and PCA, the methodology captures a broader spectrum of features related to the EEG signals, allowing for a more nuanced understanding of the data. The classifiers are tested on various combinations of features, enabling researchers to identify the most effective models for distinguishing between stress and non-stress conditions. This iterative process of evaluation and refinement aims to improve the robustness of the predictive model, ensuring that it can generalize well to unseen data. Ultimately, the outlined framework not only contributes to advancing the field of stress detection through EEG analysis but also provides a replicable blueprint for similar applications in other areas of biomedical signal processing and machine learning.

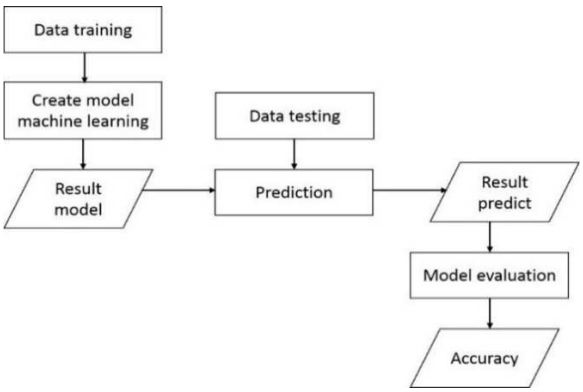

Fig. 2 Research stages

The diagram illustrates a systematic workflow for machine learning model development, encompassing the key stages of data training, testing, prediction, and evaluation. Initially, the process begins with data training, where a machine learning model is created based on a labeled dataset, allowing the model to learn from the underlying patterns within the data. Following this, the trained model undergoes data testing, which involves applying the model to new, unseen data to generate predictions. The prediction phase outputs the model's predictions, which are then compared against actual results. This comparison leads to model evaluation, where various metrics, including accuracy, are calculated to assess the model's performance. The ultimate goal is to ensure that the model performs well on the training data and generalizes effectively to new data, thereby providing reliable predictions in practical applications.

$$Accuracy = \frac{TP+TN}{TP+FP+FN+TN} \quad (4)$$

A total of 40 subjects participated, each completing three trials, with each trial lasting several seconds. To facilitate detailed analysis, each trial was segmented into 100 smaller parts. This segmentation process is crucial as it allows for a more granular examination of the EEG data, enabling researchers to extract relevant features effectively from each segment. By analyzing smaller portions of data, the models can capture transient patterns and fluctuations in the EEG signals that may be indicative of stress levels. Furthermore, this approach not only enhances the accuracy and reliability of the feature extraction process but also contributes to the overall performance of the stress classification models. By breaking the data into manageable segments, the study improves its ability to identify and distinguish between varying stress levels in the participants, ultimately leading to more precise predictions and a better understanding of the relationship between EEG signals and stress responses.

This segmentation method also ensures that even small variations in EEG signals, which might otherwise be overlooked, are captured and analyzed. By dividing the data into smaller segments, the model can detect subtle shifts in brainwave patterns associated with stress. This finer granularity enhances the model's sensitivity and accuracy in distinguishing between high and low stress levels, leading to more precise classification outcomes. Consequently, the segmentation process plays a crucial role in maximizing the efficacy of the EEG-based stress detection system.

TABLE I
ICA AND SMOTE FEATURE RESULT DATA

ICA	Label Trial 1	Result in SMOTE Trial 1	Label Trial 2	Result in SMOTE Trial 2	Label Trial 3	Result in SMOTE Trial 3
High	1000	3000	400	3600	1100	2900
Low	3000	3000	3600	3600	2900	2900
Total	4000	6000	4000	7200	4000	5800

TABLE II
PCA AND SMOTE FEATURE RESULT DATA

PCA	Label Trial 1	Result in SMOTE Trial 1	Label Trial 2	Result in SMOTE Trial 2	Label Trial 3	Result in SMOTE Trial 3
High	320	960	128	1152	352	928
Low	960	960	1152	1152	928	928
Total	1280	1920	1280	2304	1280	1856

III. RESULTS AND DISCUSSION

Related Work Research Electroencephalogram (EEG) data has been widely studied to identify a person's stress level or mental health condition. 40 subjects' EEG datasets from a variety of tasks, including the Stroop color-word test, arithmetic problem solving, identifying symmetric mirror images, and relaxing, are provided [13]. Research continues to be developed related to removing noise and artifacts from EEG dataset recordings [14], [15], [16], [17], [18], [19], [20],

[21], [22]. Electroencephalogram (EEG) signals quantify the voltage variation brought on by ionic current flowing through brain neurons. To record the electrical activity of the brain, a series of electrodes must be placed on the human scalp to collect EEG signals (See Table 3).

TABLE III
EXPERIMENTAL SCENARIO

Items	Data EEG Stroop color-word test Trial 1	Data EEG Stroop color-word test Trial 2	Data EEG Stroop color-word test Trial 3
Original ICA Features	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times
Feature ICA SMOTE	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times
ICA feature (column reduction to 16 columns)	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times
Original PCA Features	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times
Feature PCA SMOTE	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times
PCA feature (column reduction to 16 columns)	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times	Testing and evaluation of machine learning models (Decision Tree, *k*-NN, Naive Bayes, SVM, Ensemble) 3 times

Since the scalp is where the ionic current produced by the brain is measured, obstructions like the skull significantly reduce the signal quality [23]. Muscle artifacts can quickly contaminate electroencephalogram (EEG) signals, leading to incorrect interpretations in various medical diagnoses and brain-computer interface (BCI) systems. Signals from an electroencephalogram (EEG) show a person's brain's electrical activity. Even when asleep, impulses between brain cells interact. EEG data includes transitory elements, spikes, and various artifacts from the subject's movement, blinking of the eyes, and anxiety during EEG collection [10]. One method to reduce brain artifacts in EEG signals is variants of independent component analysis (ICA). The most popular method for reducing muscle artifacts in electroencephalographic signals is to partition the signals to differentiate artifactual from neural causes linearly [24]. EEG signals are frequently tainted by artifacts from the electroencephalogram (EEG), such as eyeblink, eye movement, and muscle movements. These undesired artifacts tamper with the information in the EEG signals, impairing both EEG-based brain-computer interfaces (BCIs) and qualitative analysis of therapeutic applications [25].

Numerous machine learning techniques have been proposed to build automated illness detection methods using electroencephalography (EEG) signals. EEG signals have been widely used to diagnose brain diseases such as epilepsy, Parkinson's disease (PD), and multiple sclerosis (MS) [26]. To achieve superior brain-computer interface (BCI) and human-computer interaction (HCI) communication, emotion recognition has gained attention in recent years. A crucial stage in EEG-based emotion identification is feature extraction from EEG signals.

Currently, one of the main areas of research is appropriate feature selection for EEG-based automatic emotion recognition [27], [28], [29]; One of the EEG data feature extractions is the sample entropy value and graph entropy of the EEG data [11]. Feature extraction based on time domain can provide more relevant information for classification, one of which is by using discrete wavelet transform (DWT) [30], [31], [32]. In Automatic Speech Recognition (ASR), one of the most used feature extraction methods is the Mel Frequency Cepstral Coefficient (MFCC) [33]. Selecting the right features can significantly increase the classifier's performance. In supervised machine learning techniques, useful features are extracted from the provided data by applying feature evaluation criteria once the corrected class labels of the input data are known. The selection of pertinent features and the reduction of feature dimension are the two main goals of the feature selection procedure [2].

The results of this research show that the PCA feature extraction method consistently outperforms ICA in classifying stress levels using various machine learning models. The highest accuracy was achieved using the Ensemble method, with a classification accuracy of 0.770 for PCA features and 0.745 for ICA features. The K-NN method also demonstrated strong performance, achieving an accuracy of 0.732 with ICA features and 0.700 with PCA. Furthermore, balancing the target class using the SMOTE method improved the performance of several models, particularly Naive Bayes and SVM, which saw an increase in accuracy. Reducing the number of EEG channels to 16 through correlation calculations resulted in a slight decrease in accuracy but improved computational efficiency. These findings emphasize the significance of feature extraction techniques and class balancing in improving stress classification, with PCA and Ensemble methods proving to be the most effective combination. The results suggest that refining feature selection and balancing methods can further improve model performance in EEG-based stress detection systems.

The Decision Tree method, although not as accurate as Ensemble or K-NN, still provided moderate performance,

particularly with PCA features, achieving an accuracy of 0.681. The use of ICA features generally yielded lower accuracy across all models compared to PCA, suggesting that PCA is more effective in capturing the variance in EEG data relevant to stress classification. Additionally, incorporating SMOTE for class balancing proved beneficial, particularly in addressing the imbalance between high- and low-stress levels, resulting in more reliable classification results. The reduction of EEG channels from 32 to 16 also contributed to faster processing times without significantly compromising accuracy, which is essential for real-time stress monitoring applications. Overall, the research demonstrates that a hybrid approach, combining PCA feature extraction with advanced machine learning models, especially Ensemble methods, is optimal for accurately identifying stress levels in EEG-based systems.

Based on testing and evaluation experiments, the best machine learning model is the Ensemble method, as evidenced by Tables 4, 5, 6, 7, 8, and 9, the average accuracy in both EEG Stroop color-word test trials 1, 2, and 3 data on the ICA and PCA features has the highest accuracy of 0.745 to 0.770. The best feature for identifying stress levels (low=2 and high=1) is the PCA feature based on the average accuracy values from Tables 4, 5, 6, 7, 8, and 9. Research [37] PCA feature can improve 10% accuracy in identifying stress in SAM-40 EEG data [13]. The EEG data were acquired from 40 people (mean age 21.5 years; 26 males and 14 females) in the SAM 40 dataset [1].

TABLE IV
FIRST SCENARIO ACCURACY RESULTS (PCA)

Feature PCA (Method)	EEG Stroop color-word test Trial 1	EEG Stroop color-word test Trial 2	EEG Stroop color-word test Trial 3
Decision Tree	0.634	0.792	0.596
K-NN	0.618	0.828	0.591
Naive Bayes	0.755	0.896	0.715
SVM	0.753	0.901	0.723
Ensemble	0.695	0.889	0.650

TABLE V
SECOND SCENARIO ACCURACY RESULTS (ICA)

Feature ICA (Method)	EEG Stroop color-word test Trial 1	EEG Stroop color-word test Trial 1	EEG Stroop color-word test Trial 1
Decision Tree	0.678	0.801	0.645
K-NN	0.797	0.855	0.802
Naive Bayes	0.553	0.596	0.537
SVM	0.533	0.550	0.514
Ensemble	0.724	0.892	0.700

TABLE VI.
SECOND SCENARIO ACCURACY RESULTS (PCA)

Feature PCA (Method)	EEG Stroop color-word test Trial 1	EEG Stroop color-word test Trial 2	EEG Stroop color-word test Trial 3
Decision Tree	0.674	0.807	0.634
K-NN	0.691	0.795	0.677
Naive Bayes	0.608	0.770	0.586
SVM	0.497	0.537	0.432
Ensemble	0.766	0.937	0.756

TABLE VII.
COLUMN REDUCTION RESULTS

Feature	EEG Stroop color-word test Trial 1	EEG Stroop color-word test Trial 2	EEG Stroop color-word test Trial 3
ICA	1, 2, 3, 6, 8, 9, 10, 11, 14, 16, 17, 19, 22, 27, 30, 32	1, 3, 8, 9, 11, 12, 16, 17, 18, 20, 24, 25, 27, 28, 30, 31	2, 4, 7, 10, 12, 14, 15, 17, 19, 22, 23, 24, 27, 29, 30, 32
PCA	1, 3, 5, 6, 7, 10, 13, 14, 15, 16, 18, 19, 20, 24, 30, 32	1, 3, 7, 8, 11, 12, 13, 18, 21, 22, 23, 24, 25, 29, 30, 32	1, 2, 3, 6, 7, 10, 14, 16, 17, 18, 26, 27, 29, 30, 31, 32

TABLE VIII
RESULTS OF THE ACCURACY OF THE THIRD SCENARIO (ICA)

Feature ICA (Method)	EEG Stroop color-word test Trial 1	EEG Stroop color-word test Trial 1	EEG Stroop color-word test Trial 1
Decision Tree	0.640	0.820	0.590
K-NN	0.651	0.815	0.608
Naive Bayes	0.755	0.899	0.708
SVM	0.755	0.899	0.711
Ensemble	0.683	0.872	0.640

TABLE IX
RESULTS OF THE ACCURACY OF THE THIRD SCENARIO (PCA)

Feature PCA (Method)	EEG Stroop color-word test Trial 1	EEG Stroop color-word test Trial 2	EEG Stroop color-word test Trial 3
Decision Tree	0.605	0.805	0.585
K-NN	0.629	0.819	0.651
Naive Bayes	0.755	0.884	0.743
SVM	0.757	0.884	0.743
Ensemble	0.704	0.867	0.668

In this research, all forty subjects were included, a 32-channel Emotiv Epoc Flex gel kit (Emotiv Inc., San Francisco, CA, USA) was used to collect the EEG data, and samples were taken at a rate of 128 per second. The patient's EEG signals were captured throughout a variety of tasks, including arithmetic problem-solving, the Stroop color-word test, identifying symmetric mirror images, and relaxing. Every single job was completed in twenty-five seconds. For every job, three trials were recorded. After processing, the EEG data were cleaned of artifacts. Based on the degree of stress they encountered during a certain task in a trial, each participant gave comments on a scale from 1 to 10. Table 10 explains that using the features: frequency domain (BP), time domain (HP), non-linear features (HFD), and raw EEG data provides good accuracy above 75 [43]. The results of the accuracy for the ICA features show varying performances across the different machine learning models.

TABLE X
SAM 40 EEG DATA ACCURACY RESULTS

Feature	Classification Methods	Accuracy
Frequency Domain (BP)	SVM	87.667
Time Domain (HP)	SVM	86.500
Non-linear Features (HFD)	SVM	89.830
Raw EEG	EEGNet	92.500
Raw EEG	Shallow ConvNet	90.440
Raw EEG	Deep ConvNet	75.750

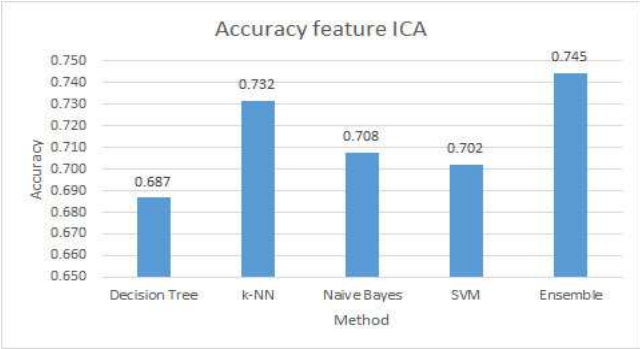

Fig. 3 Graph Accuracy feature ICA

The Decision Tree model achieved an accuracy of 0.687, indicating moderate performance in classifying stress levels based on ICA features. The K-NN (K-Nearest Neighbors) model outperformed the others with an accuracy of 0.732, suggesting it was the most effective at leveraging the ICA features. The Naive Bayes method followed closely with an accuracy of 0.708, showing its capability in handling this classification task. The SVM (Support Vector Machine) model achieved an accuracy of 0.702, which was slightly lower than Naive Bayes but still relatively close. Finally, the Ensemble method demonstrated the highest accuracy at 0.745, indicating that combining multiple models resulted in the best performance for stress classification based on ICA features.

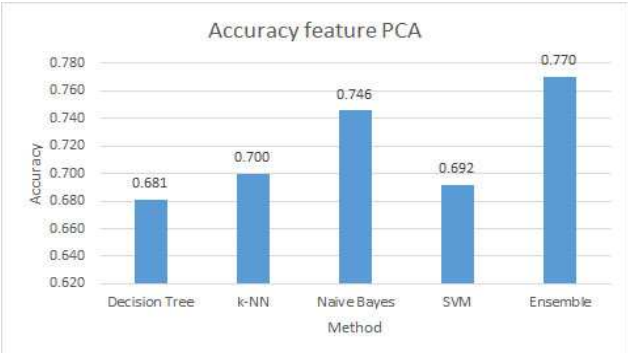

Fig. 4 Accuracy comparison value

The results of the PCA feature accuracy comparison highlight different performances among the machine learning models. The Decision Tree model obtained an accuracy of 0.681, showing moderate performance in classifying stress levels using PCA features. The K-NN (K-Nearest Neighbors) model performed slightly better, with an accuracy of 0.700. The Naive Bayes method, however, showed significant improvement with an accuracy of 0.746, indicating its effectiveness in handling PCA-based features. The SVM (Support Vector Machine) model followed with an accuracy of 0.692, showing slightly lower performance than k-NN. Finally, the Ensemble method outperformed all other models with the highest accuracy of 0.770, demonstrating that combining multiple models was the most effective approach for stress classification when using PCA features.

IV. Conclusion

This research identifies stress levels (low=2, and high=1) based on ICA and PCA features using various machine learning methods, including Decision Tree, K-NN, Naive Bayes, SVM, and Ensemble. The study analyzes both the most effective features and machine learning models for stress level classification. Based on the experiment results, PCA emerges as the best feature for stress identification, achieving an average accuracy value of 0.718. Among the machine learning models tested, the Ensemble method consistently outperformed the others, achieving sequential accuracy values of 0.745 for ICA features and 0.770 for PCA features. These findings highlight the potential of PCA and Ensemble techniques in enhancing the accuracy and reliability of stress classification systems. Future research could explore further optimization of these models or apply the approach to a broader range of datasets to validate its generalizability.

Nomenclature

EEG	Electroencephalogram
ICA	Independent Component Analysis
PCA	Principal Component Analysis
SVM	Support Vector Machine
K-NN	K-Nearest Neighbors
Ensemble	Ensemble Learning Model
SAM	Self-Assessment Manikin
DT	Decision Tree
NB	Naïve Bayes

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